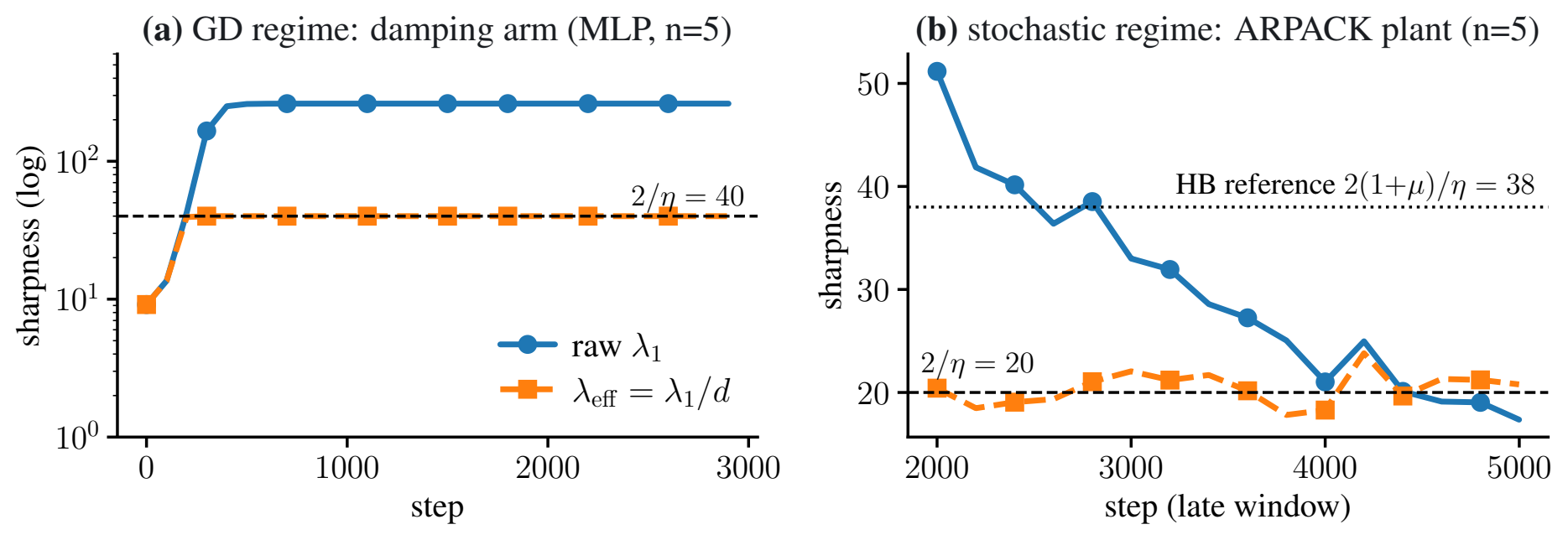

(a) GD regime: damping arm (MLP, n=5)
sharpness (log)
10^2
10^1
10^0
$2/\eta = 40$
raw λ_1
$\lambda_{\text{eff}} = \lambda_1/d$
0
1000
2000
3000
step
(b) stochastic regime: ARPACK plant (n=5)
sharpness
50
40
30
20
HB reference $2(1+\mu)/\eta = 38$
$2/\eta = 20$
2000
3000
4000
5000
step (late window)